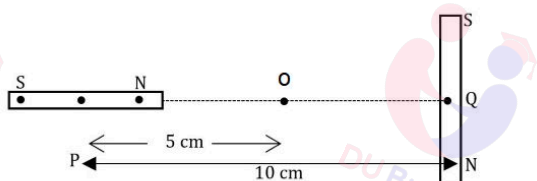


Section : Physics

Q.1

Two identical short bar magnets each of magnetic moment 12.5 Am^2 are placed with 10 cm separation between their centers as shown in the figure. The magnitude of magnetic field at O is



1. $\sqrt{5} \times 10^{-2} \text{ T}$
2. $5 \times 10^{-2} \text{ T}$
3. $\sqrt{3} \times 10^{-2} \text{ T}$
4. $3 \times 10^{-2} \text{ T}$

Options 1. 1

2. 2
3. 3
4. 4

Q.2

'n' identical small drops are charged to V volt each. They coalesce to form a bigger drop. The potential of the bigger drop will be:

1. V
2. nV
3. $n^{\frac{2}{3}}V$
4. $n^{\frac{1}{3}}V$

Options 1. 1

2. 2
3. 3
4. 4

Q.3

Two sources of equal emf are connected in series. This combination is connected to an external resistance R.

The internal resistance of two sources are r_1 and r_2 ($r_2 > r_1$).

If the potential difference across the source of internal resistance r_2 is zero, then R equals to:

1. $R = r_2 - r_1$
2. $R = r_1 - r_2$
3. $R = r_1$
4. $R = 0$

Options 1. 1

2. 2
3. 3
4. 4

Q.4 Match List-I with List-II

List-I

List-II

Electromagnetic Waves Production

- | | |
|-------------------------|---|
| (A) Radio waves | (I) Vibration of atoms and molecules |
| (B) X-rays | (II) originates from an atomic nucleus |
| (C) γ -radiation | (III) produced by accelerating electrons in a circuit |
| (D) infrared waves | (IV) emitted from heavy atoms |

Choose the correct answer from the options given below:

1. (A) - (IV), (B) - (II), (C) - (III), (D) - (I)
2. (A) - (II), (B) - (III), (C) - (I), (D) - (IV)
3. (A) - (III), (B) - (I), (C) - (IV), (D) - (II)
4. (A) - (III), (B) - (IV), (C) - (II), (D) - (I)

Options 1. 1

2. 2

3. 3

4. 4

- Q.5** When a charged particle moves in a uniform magnetic field, its kinetic energy:
1. goes on increasing
 2. may increase or decrease depending on angle between its velocity and magnetic field
 3. remains unchanged
 4. may increase or decrease depending upon the sign of charge

Options 1. 1

2. 2

3. 3

4. 4

- Q.6** Arrange the following radiations in the decreasing order of their energy

- (A) Microwaves
(B) γ -rays
(C) AM radio waves
(D) Infrared waves

Choose the correct answer from the options given below:

1. (B), (D), (A), (C)

2. (A), (B), (D), (C)

3. (B), (A), (D), (C)

4. (C), (B), (D), (A)

Options 1. 1

2. 2

3. 3

4. 4

- Q.7** A proton and an alpha particle moving with equal kinetic energies enter perpendicularly into a magnetic field.

If r_p and r_α are the respective radii of their circular paths, what will be the ratio $\frac{r_p}{r_\alpha}$?

1. $1 : \sqrt{2}$

2. $\sqrt{2} : 1$

3. $1 : 1$

4. $1 : 2$

Options 1. 1

2. 2

3. 3

4. 4

- Q.8** The smallest wavelength in the spectral lines of Paschen series in the hydrogen atom is
(Given: $R = 1.097 \times 10^7 \text{ m}^{-1}$)

1. 4102 \AA

2. 6304 \AA

3. 8204 \AA

4. 9846 \AA

Options 1. 1

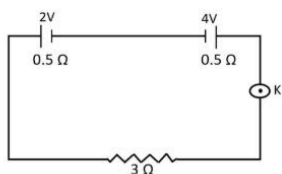
2. 2

3. 3

4. 4

Q.9

Two cells of emf 2 V and 4 V and internal resistance of $0.5\ \Omega$ each are connected with an external resistor of resistance $3\ \Omega$ as shown in the figure. The terminal potential difference of 2 V cell is:



1. 2.25 V
2. 1.75 V
3. 1.5 V
4. 2.5 V

Options 1. 1

2. 2
3. 3
4. 4

Q.10

A neutron is absorbed by a ${}^6_3\text{Li}$ nucleus with subsequent emission of an alpha particle. The energy released in the reaction is

Use mass of ${}^6_3\text{Li} = 6.015126\ \text{u}$

Mass of ${}^4_2\text{He} = 4.002604\ \text{u}$

Mass of ${}_1^1\text{n} = 1.008665\ \text{u}$

Mass of ${}_1^3\text{H} = 3.016049\ \text{u}$ and $1\ \text{u} = 931\ \text{MeV}$

1. 4.78 MeV
2. 5.28 MeV
3. 3.44 MeV
4. 5.58 MeV

Options 1. 1

2. 2
3. 3
4. 4

Q.11

A light bulb and an air core inductor are connected to an AC source. The glow of the light bulb increases if

- (A) the frequency of an AC source increases
- (B) a copper rod is inserted inside the inductor
- (C) the frequency of an AC source decreases
- (D) an iron rod is inserted inside the inductor

Extra close brace or missing open brace

Choose the correct answer from the options given below:

1. (A), (B) and (E) only
2. (B), (C) and (D) only
3. (C), (D) and (E) only
4. (B), (C) and (E) only

Options 1. 1

2. 2
3. 3
4. 4

Q.12

Two charged conducting spheres of radii a and b are connected by a wire. The ratio of the electric fields at their surfaces respectively, is

1. $a:b$
2. $ab:1$
3. $b:a$
4. $1:ab$

Options 1. 1

2. 2
3. 3
4. 4

Q.13

A $400\ \text{pF}$ capacitor is charged by a $200\ \text{V}$ supply. It is then disconnected from the supply and is connected to another

uncharged 400 pF capacitor. The electrostatic energy lost in the process is

1. $2.4 \times 10^{-6} \text{ J}$
2. $4 \times 10^{-6} \text{ J}$
3. $6.0 \times 10^{-6} \text{ J}$
4. $8.0 \times 10^{-6} \text{ J}$

Options 1. 1

2. 2
3. 3
4. 4

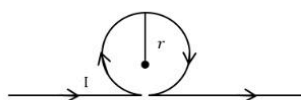
- Q.14** In a full wave rectifier, the output voltage obtained is in the shape of pulsating half sinusoids which are unidirectional but not steady. To get a steady d.c. output
- (A) An inductor can be use in series with the load resistor
(B) A capacitor of high value of capacitance can be used in parallel with the load resistor
(C) A capacitor of low capacitance can be used in series with the load resistor
(D) Only load resistor is enough to filter out the pulses and convert them to d.c.
- Choose the correct answer from the options given below:
1. (C) and (D) only
 2. (A) and (B) only
 3. (A), (B) and (D) only
 4. (D) only

Options 1. 1

2. 2
3. 3
4. 4

Q.15

A part of an infinitely long current carrying wire is bent into a circular shape as shown in the diagram. The magnetic field at the centre of the circle is



1. $\frac{\mu_0 I}{2\pi r} (\pi + 1)$ normally inward
2. $\frac{\mu_0 I}{2\pi r} (\pi + 1)$ normally outward
3. $\frac{\mu_0 I}{2\pi r} (\pi - 1)$ normally outward
4. $\frac{\mu_0 I}{2\pi r} (\pi - 1)$ normally inward

Options 1. 1

2. 2
3. 3
4. 4

Q.16

The intensity ratio of the two interfering beams of light is β . The value of $\left[\frac{I_{\max} - I_{\min}}{I_{\max} + I_{\min}} \right]$ is

1. $2\sqrt{\beta}$
2. $\frac{2\sqrt{\beta}}{(1 + \beta)}$
3. $\frac{2}{(1 + \beta)}$
4. $\frac{(1 + \beta)}{2\sqrt{\beta}}$

Options 1. 1

2. 2
3. 3
4. 4

Q.17 A 100 W bulb working on 200 V and a 200 W bulb on 100 V, have their resistances in the ratio

- 1) 4:1
- 2) 8:1
- 3) 1:2
- 4) 2:1

Options 1. 1

2. 2
3. 3
4. 4

Q.18 The kinetic energy of an electron in the H-atom in the second excited state is k units. Its total energy and potential energy, respectively, in the ground state will be

1. $-9k$, $-18k$
2. $-k$, $-2k$
3. $-4k$, $-16k$
4. $-k$, $2k$

Options 1. 1

2. 2
3. 3
4. 4

Q.19 Arrange the range of frequencies of the following series of hydrogen atom in ascending order

- (A) Balmer
- (B) Lyman
- (C) Paschen
- (D) Brackett

Choose the correct answer from the options given below:

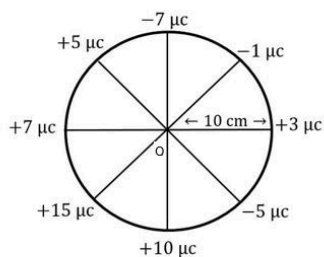
1. (B), (A), (C), (D)
2. (B), (A), (D), (C)
3. (C), (D), (A), (B)
4. (D), (C), (A), (B)

Options 1. 1

2. 2
3. 3
4. 4

Q.20

Eight charges having values as shown in figure are arranged symmetrically on a circle of radius 10 cm in air. The potential at the centre O is



1. $2.43 \times 10^6 \text{ V}$
2. $24.3 \times 10^6 \text{ V}$
3. $243 \times 10^6 \text{ V}$
4. $38.6 \times 10^6 \text{ V}$

Options 1. 1

2. 2
3. 3
4. 4

Q.21 Light from a point source in the air falls on a convex spherical glass surface having radius of curvature 20 cm and refractive index 1.5. The distance of the light source from the glass surface is 100 cm. The distance of the image from the glass surface is:

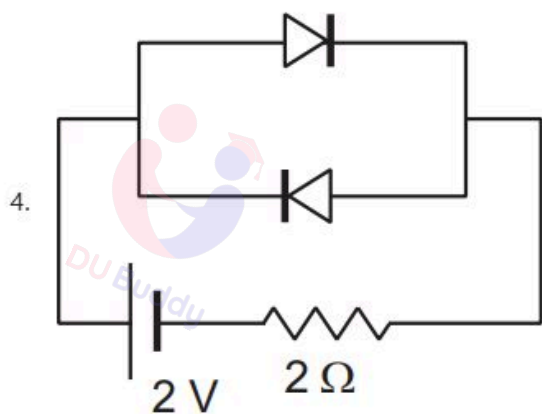
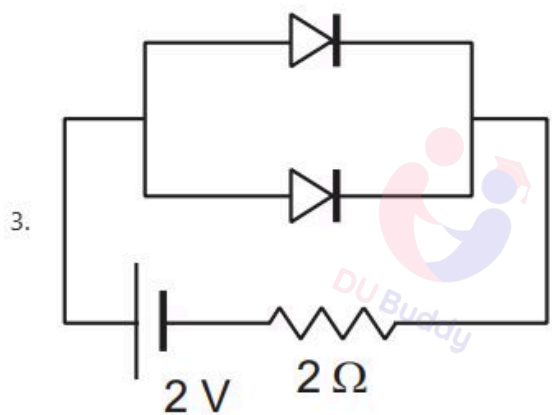
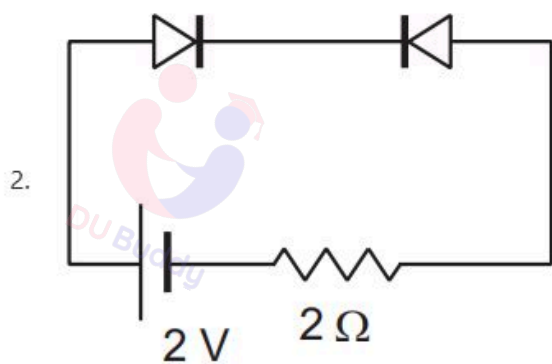
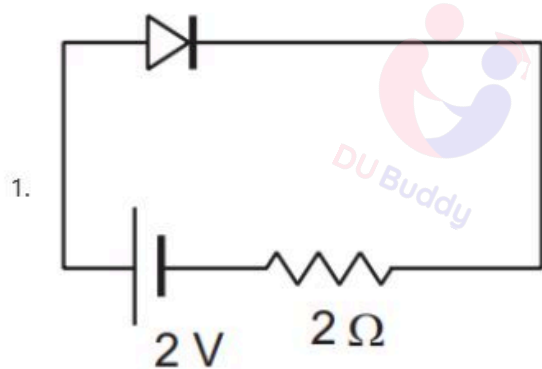
1. 50 cm
2. 70 cm
3. 100 cm
4. 200 cm

Options 1. 1

2. 2

Q.22

Which of the following will give zero current through the resistor of $2\ \Omega$.



Options 1. 1

2. 2

3. 3

4. 4

Q.23 In a hydrogen atom, an electron moves in an orbit of radius 5×10^{-11} m with a speed of 2.2×10^6 m/s. The equivalent current is

1. 1.12×10^{-3} A
2. 3.36×10^{-3} A
3. 5 mA
4. $7 \mu\text{A}$

Options 1. 1

2. 2
3. 3
4. 4

Q.24 When light of suitable frequency is incident on a photosensitive material, photo electrons are ejected and photoelectric current flows. If the intensity of the incident light is increased, then

1. The maximum kinetic energy of photoelectrons will increase
2. The stopping potential will increase
3. The stopping potential will decrease
4. The saturation current will increase

Options 1. 1

2. 2
3. 3
4. 4

Q.25 A long solenoid with 25 turns per cm has a small loop of area 2.0 cm^2 placed inside the solenoid normal to its axis. If the current carried by the solenoid changes steadily from 3.0 A to 5.0 A in 0.1 s. The induced emf in the loop due to change in the current is

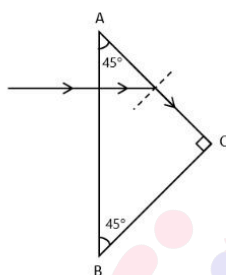
1. $4\pi \times 10^{-8}$ V
2. $4\pi \times 10^{-7}$ V
3. $4\pi \times 10^{-6}$ V
4. $4\pi \times 10^{-4}$ V

Options 1. 1

2. 2
3. 3
4. 4

Q.26

When a ray of light is incident on the face AB of prism normally, after refraction it is found to graze the side AC as shown in the figure, the refractive index of the prism is



1. 1.5
2. 1.414
3. 1.33
4. 1.73

Options 1. 1

2. 2
3. 3
4. 4

Q.27 Two point charges $3 \mu\text{C}$ and $-3 \mu\text{C}$ are located at points A : (3 cm, 0, 0) and B : (-3 cm, 0, 0), respectively. The electric field at the origin is

1. $+6 \times 10^7 \hat{i} \text{ N/C}$
2. $+6 \times 10^7 \hat{j} \text{ N/C}$
3. $-6 \times 10^7 \hat{i} \text{ N/C}$
4. 0

Options

1. 1

2. 2
3. 3
4. 4

Q.28 Conditions for obtaining a interference pattern of light are

- (A) Two waves should be coherent
- (B) The distance of the screen from the slits should be small
- (C) The separation between slits should be small
- (D) Two coherent waves should have nearly the same amplitude

Choose the correct answer from the options given below:

1. (A), (B) and (C) only
2. (B), (C) and (D) only
3. (A), (B) and (D) only
4. (A), (C) and (D) only

Options

1. 1

2. 2
3. 3
4. 4

Q.29 You are sitting in a parked car and you notice a boy running towards the car in the side view mirror of the car. If the

boy is running at a constant speed towards you, which option according to you is correct

1. The boy will appear to be running at a constant speed towards you
2. The boy will appear to be stationary
3. The boy will appear to be running towards the car and his speed will appear to be continuously increasing
4. The boy will appear to be running towards the car and his speed will appear to be continuously decreasing

Options

1. 1

2. 2
3. 3
4. 4

Q.30 A proton has double the kinetic energy of an α - particle. The ratio of the de-Broglie wavelength of the proton to that of α - particle will be

1. $\frac{1}{2}$
2. $\frac{1}{\sqrt{2}}$
3. $\frac{\sqrt{2}}{1}$
4. $\frac{2}{1}$

Options

1. 1

2. 2
3. 3
4. 4

Q.31

A circuit containing a $\frac{2}{\pi}$ H inductor, a $\frac{40}{\pi}$ μ F capacitor and a 50Ω resistor in series are connected to an a.c. source of emf $v = 230 \sin 314t$. The amplitude of the current in the circuit is

1. 3.25 A
2. 4.6 A
3. 5.8 A
4. 6.2 A

Options 1. 1

2. 2
3. 3
4. 4

Q.32

A cell of constant emf is first connected to a resistance R_1 and then connected to a resistance R_2 . If power delivered in both cases is same, then the internal resistance of the cell is

1. $\sqrt{R_1 R_2}$
2. $\sqrt{\frac{R_1}{R_2}}$
3. $\frac{R_1 - R_2}{2}$
4. $\frac{R_1 + R_2}{2}$

Options 1. 1

2. 2
3. 3
4. 4

Q.33

A galvanometer shows full scale deflection for a current of 4 mA. It is converted into an ammeter of range from 0 to 8 A by using a shunt. The ratio of resistance of the galvanometer coil to the resistance of the shunt used is nearly:

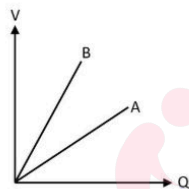
1. 2000 : 1
2. 1:2000
3. 1000 : 1
4. 1:1000

Options 1. 1

2. 2
3. 3
4. 4

Q.34

The graph shows the variation of potential difference between the plates of two capacitors A and B with the increase of charge stored by them. From the graph we can conclude:



1. Capacitance of A is greater than B
2. Capacitance of B is greater than A
3. B stores more electrostatic energy than A at a given potential
4. A stores more electrostatic energy than B at a given charge

Options 1. 1

2. 2
3. 3
4. 4

Q.35

A series LCR circuit with $L = 5.0$ H, $C = 80 \mu$ F and $R = 40 \Omega$ is connected to a variable frequency 200 V source. The amplitude of current at the resonant frequency is

1. 5.0 A
2. 6.2 A
3. 7.1 A
4. 8.4 A

Options 1. 1

2. 2
3. 3
4. 4

Q.36 Six identical cells, each of emf 3 V and internal resistance of r , are connected in series to a $3\ \Omega$ resistance. If the net current flowing in the circuit is 2.0 A. The internal resistance of the cell is

1. $\frac{1}{2}\ \Omega$
2. $\frac{1}{4}\ \Omega$
3. $\frac{1}{3}\ \Omega$
4. $1\ \Omega$

Options 1. 1

2. 2
3. 3
4. 4

Q.37 A field of 2×10^3 A/m produces a flux of 2.4×10^{-5} Wb in an iron bar having area of cross-section 0.2 cm^2 . The relative permeability of medium is:

1. 478
2. 380
3. 600
4. 750

Options 1. 1

2. 2
3. 3
4. 4

Q.38 Daily life phenomena based on scattering of light are

- (A) Blue colour of sky
- (B) Clouds appearing white
- (C) Twinkling of stars
- (D) Danger signals being red

Choose the correct answer from the options given below:

1. (A), (B) and (D) only
2. (A), (B) and (C) only
3. (A), (B), (C) and (D)
4. (B), (C) and (D) only

Options 1. 1

2. 2
3. 3
4. 4

Q.39 In an interference experiment, the two slits are 1 mm apart and the screen is placed 1 m away from the slits. The width of a bright fringe for the light of wavelength 500 nm is

1. 0.25 mm
2. 5.0 mm
3. 0.5 mm
4. 2.5 mm

Options 1. 1

2. 2
3. 3
4. 4

Q.40 In transmission of power from generation plants to users, transformers are used. Identify the correct reason to choose appropriate transformers.

- (A) A step-up transformer is used at the generation plant and high voltages are transmitted.
 (B) A step-down transformer is used at the generation plant and low voltages are transmitted.
 (C) A step-up transformer lowers the current and hence loss due to joule heating is less.
 (D) A step-down transformer lowers the current and hence loss due to joule heating is less.

Choose the correct answer from the options given below:

1. (A) and (C) only
2. (A) only
3. (B) and (D) only
4. (D) only

Options

1. 1
2. 2
3. 3
4. 4

Q.41 In an LC oscillator, $C = 50 \mu\text{F}$ and $L = 0.08 \text{ H}$ are used. If the resistance of the circuit is negligible, the time period of oscillation is

1. 1.256 ms
2. 6.28 ms
3. 12.56 ms
4. 0.628 ms

Options

1. 1
2. 2
3. 3
4. 4

Q.42 The energy of a photon of wavelength $1 \text{ \AA}$ is
 (Given: $h = 6.63 \times 10^{-34} \text{ J s}$)

1. $1.24 \times 10^4 \text{ eV}$
2. $2.48 \times 10^3 \text{ eV}$
3. $3.62 \times 10^4 \text{ eV}$
4. $4.8 \times 10^6 \text{ eV}$

Options

1. 1
2. 2
3. 3
4. 4

Q.43 A straight wire of mass 300 g and length 1.5 m carries a current of 2 A. It is suspended in mid air by a uniform horizontal magnetic field. The magnitude of the magnetic field is
 (Neglect the earth's magnetic field and $g = 10 \text{ m/s}^2$)

1. 3T
2. 2T
3. 1T
4. 0.5 T

Options

1. 1
2. 2
3. 3
4. 4

Q.44 Two slits are made 0.8 mm apart and the screen is placed 1.5 m away. What should be the width of each slit to obtain 8 maxima of the double slit pattern within the central maximum of the single slit pattern?

1. 0.2 mm
2. 0.4 mm
3. 0.6 mm
4. 0.8 mm

Options

1. 1
2. 2
3. 3
4. 4

Q.45 A charge q is placed at the centre of a sphere of radius r . The electric flux crossing the sphere is Φ . If both the charge

and radius of the sphere are doubled, the electric flux crossing the sphere will be

1. Φ only
2. 2Φ
3. 4Φ
4. 8Φ

Options

1. 1
2. 2
3. 3
4. 4

Q.46 Two point charges $q_1 = 10 \times 10^{-8} \text{ C}$ and $q_2 = -2 \times 10^{-8} \text{ C}$ are separated by a distance of 6 cm. The distance from q_1 where the electric potential is zero, can be

1. 5 cm only
2. 5 cm and 7.5 cm
3. 7.5 cm only
4. 5 cm and 1.5 cm

Options

1. 1
2. 2
3. 3
4. 4

Q.47 Current and voltage in an a. c. circuit are given as

$$I = 10 \sin \left(50t - \frac{\pi}{2} \right) \text{ A}$$

$$V = 200 \sin(50t) \text{ V } (t \text{ represents time})$$

The power dissipated in the circuit is

1. 2000 W
2. 20 W
3. 0 W
4. 40 W

Options

1. 1
2. 2
3. 3
4. 4

Q.48 Arrange the following materials in increasing order of their resistivities

- (A) Tungsten
- (B) Aluminium
- (C) Gold
- (D) Copper

Choose the correct answer from the options given below:

1. (A)<(B)<(C)<(D)
2. (D)< (C)<(B)<(A)
3. (B)<(A)<(D)<(C)
4. (C)<(D)<(B)<(A)

Options

1. 1
2. 2
3. 3
4. 4

Q.49 An electromagnetic wave travels in a medium at a speed of $2 \times 10^8 \text{ m/s}$. If the relative permeability (μ_r) of the medium is 1.0, then the relative permittivity (ϵ_r) of the medium is equal to

1. 1.0
2. 2.25
3. 4.51
4. 6.2

Options

1. 1
2. 2

3. 3

4. 4

Q.50 In a silicon intrinsic semiconductor, the forbidden energy gap between the valence band and the conduction band is approximately equal to

- 1. 1.1 eV
- 2. 2.7 eV
- 3. 3.3 eV
- 4. 4.0 eV

Options

- 1. 1
- 2. 2
- 3. 3
- 4. 4